CABLESSail: Solar Sail Momentum Management Using Shape Control

Soojeong Lee and Ryan Caverly
Department of Aerospace Engineering & Mechanics, University of Minnesota, Minneapolis, MN, USA

Motivation & Objectives

Goal
- Develop an actuation technology that generates large solar sail momentum management torques via shape control.

Motivation
- Larger solar sails will have large unwanted boom deformations, which results in solar radiation pressure (SRP) disturbance torques.
- Reaction wheel momentum management requires additional actuators that can create reliably-large torques in all 3 axes.

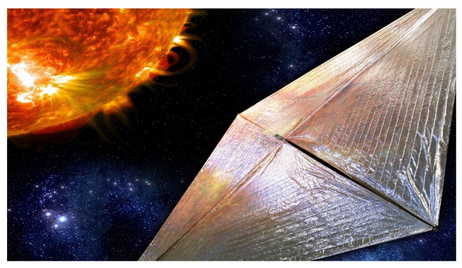


Image Credit: NASA

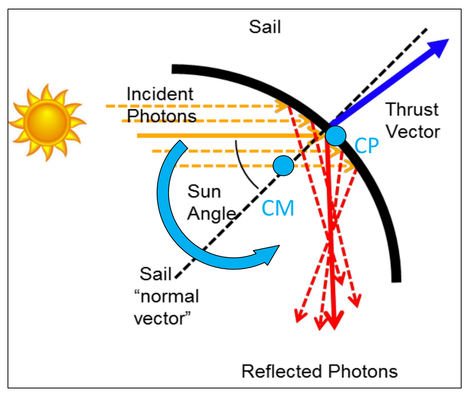
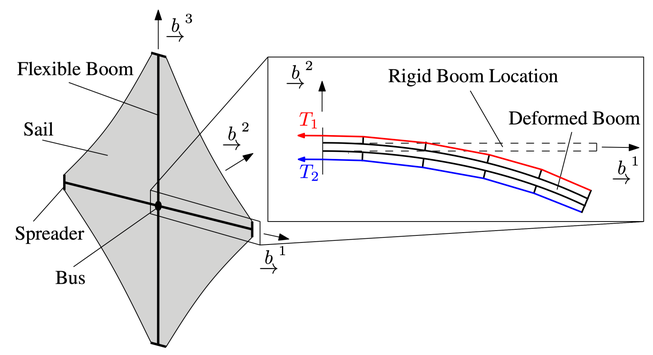


Image Credit: NASA

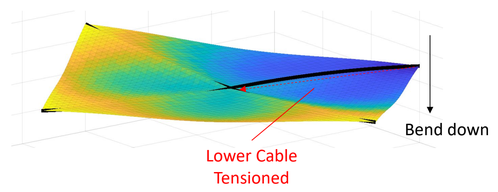
CABLESSail Concept

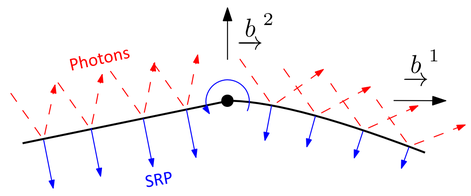
Cable-Actuated Bio-inspired Lightweight Elastic Solar Sail

- CABLESSail uses motor-actuated cables running along the solar sail's booms to precisely control the solar sail's shape.



Torque Generation
- Increased cable tension induces boom bending. This changes the local Sun incidence angle and generates a net SRP torque.

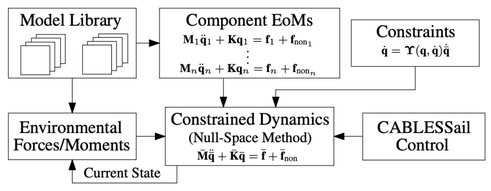




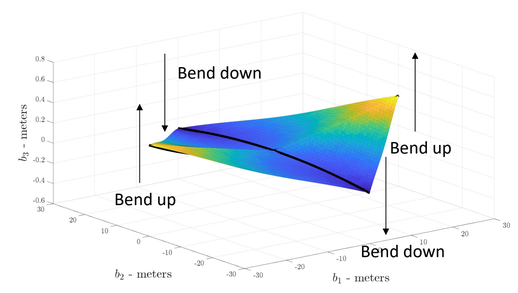
CABLESSail Simulation

CABLESSail Modeling
- Modular static and dynamic flexible multi-body simulation code that is amenable to the evaluation of different designs.
- Code is publicly available on GitHub.

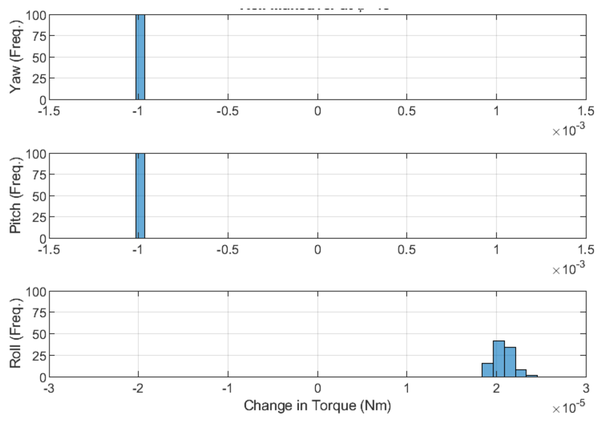
Steady-State Torques Generated Across Sail Shapes
- Ran 100 Monte Carlo simulations of Solar Cruiser.



Roll Maneuver



Change in Torque Produced by Roll Maneuver

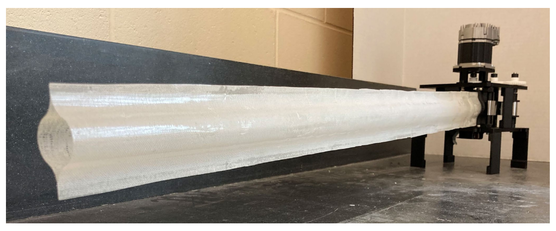


Prototype Development

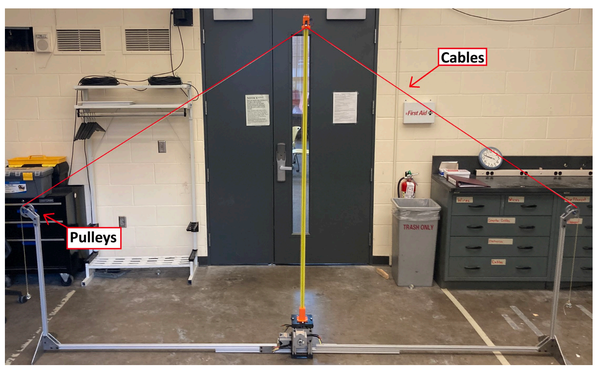
Metallic Tape Measure TRAC Boom



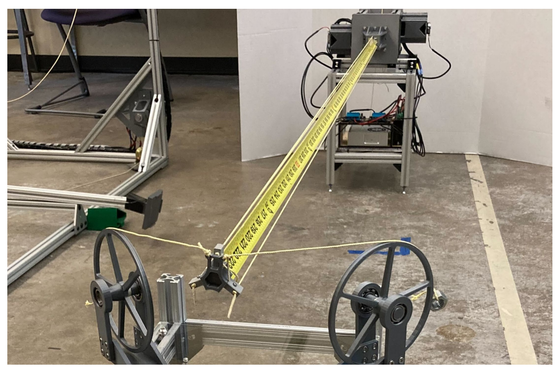
Fiberglass Composite Boom



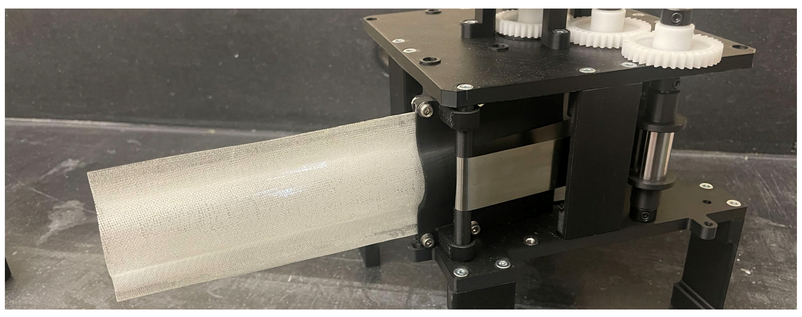
Vertical Deployed Prototype

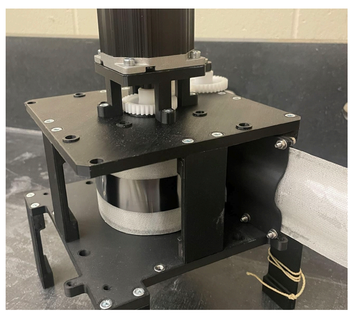


Horizontal Deployed Prototype



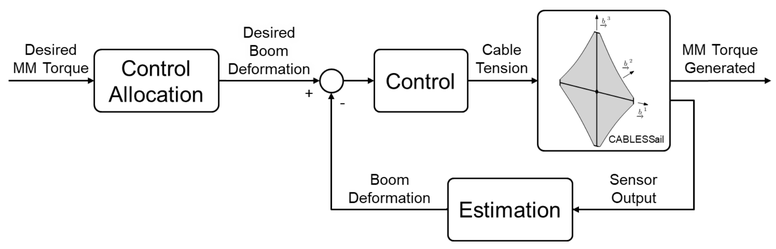
Deployable Prototype with Lenticular Composite Boom



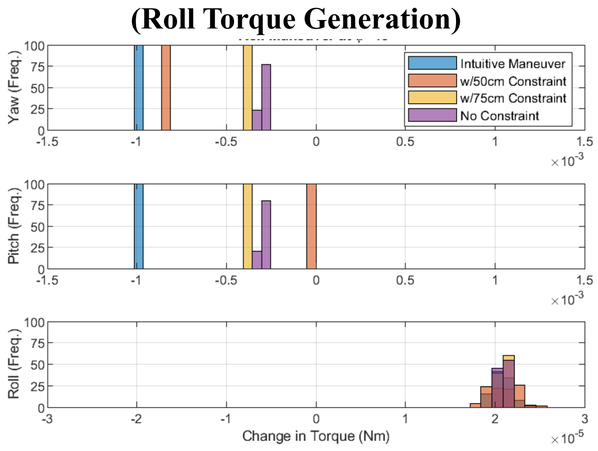


Estimation & Control

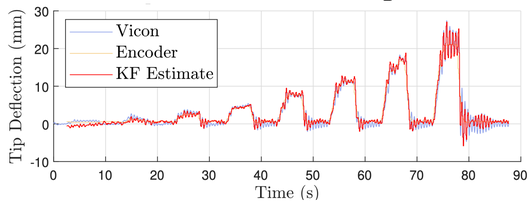
- A control allocation algorithm determines the boom tip deformations needed to generate the desired momentum management torque, then a feedback controller uses boom tip deformation estimates to track the desired deformations.



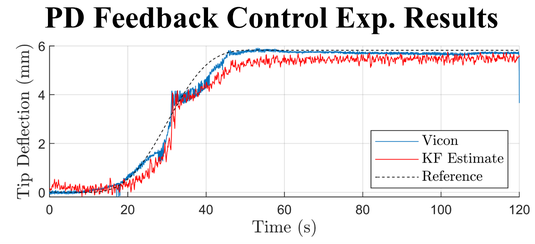
Control Allocation Simulation Results (Roll Torque Generation)



Kalman Filter Estimation Exp. Results



PD Feedback Control Exp. Results



Future Outlook & Conclusion

- CABLESSail is a promising technology for solar sail momentum management torque generation.
- Currently rated at TRL 2 with plans to achieve TRL 3 (target Oct. 2026) and beyond.
- Will investigate other use cases, such as control of drag sail.

Funding

- Early Career Faculty grant from NASA's Space Technology Research Grants Program (80NSSC23K0075)

NASA Partner

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Paper
arXiv:2512.13493

Emails: lee03939@umn.edu, rcaverly@umn.edu